

SEMINAR 1

The Proton Radius Puzzle

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Abstract

In this seminar, we start with a chronological overview of existing proton radius measurements using two entirely different experimental techniques, which both at the beginning led to the same result. However, in 2010, a new technique was focusing on the investigations of muonic hydrogen, which could alter much more precise results, but significantly differ from previous measurements. After observation of the discrepancy, a series of new measurements were performed in order to give an explanation for the inconsistency, but instead ended with two groups of results - one gathered around 0.88 fm and the other around 0.84 fm.

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1 Introduction

In 1911, Rutherford and his students performed the famous Rutherford experiment and observed that alpha particles from radioactive decays occasionally scatter at angles greater than 90° , which is possible only if they were scattering off something more massive than themselves. Rutherford then concluded that the positive charge in an atom is concentrated into a small, compact region and he was able to describe data (figure 1) with the Rutherford formula

$$\left(\frac{d\sigma}{d\Omega}\right)_{\text{Ruth.}} = \frac{Z^2\alpha^2}{4E^2 \sin^4 \frac{\theta}{2}}, \quad (1)$$

where θ is the scattering angle, E is the energy of the probe and α is the fine-structure constant. But as he increased the energy of the α particles, at some point (27.5 MeV) the data started to deviate from the formula (1). The interpretation at the time was that this is the energy where α particles start to enter the nucleus, which was the first indication that the nucleus is not a point-like structure but has a finite size.

It took another decade of experiments for Rutherford to discover that a proton is produced in α -scattering from nitrogen (air). Rutherford observed that when α -particles were shot into the air, hydrogen nuclei were detected as products. He continued with other atoms as targets and got the same byproduct, from which he concluded that the hydrogen nucleus plays a fundamental role in atomic structure and called it a proton (1920). In 1933, Otto Stern measured the magnetic moment¹ of the proton, which resulted in 2.5-times larger value than the theory predicted. This implied that the proton is not an elementary point-like particle as well.

Following scattering experiments started to replace α -particle as a probe with an electron, which predominantly interacts with electro-magnetic interaction and can therefore provide much better results. The development of scattering experiments led to the first extraction of the proton charge radius, published in 1963 [1]. Similarly, with the progress of Quantum Field Theory (QFT), the proton charge radius was obtained in 1979 [2] with a completely different method, called hydrogen spectroscopy. The basic idea is to measure the energies of bound states in hydrogen atom and compare them to theoretical calculations. Since theories include corrections for the finite size of the proton, it is possible to extract it from the measurements.

Until 2010, many experiments of the proton charge radius were performed with both methods, and the established value of 0.8775(51) fm was published as a recommended by the Committee on Data for Science and Technology (CODATA) [3].

However, in 2010, a new technique called muonic hydrogen spectroscopy was developed, giving a value of 0.84184(67) fm [4]. The result of this experiment was ten times more precise than any of the previous measurements, but at the same time differing from the accepted CODATA value by 5σ . The experiment was repeated in 2013 [5] and produced even more precise value, differing from the CODATA value by 5.2σ . The question at the time was, why is this new experiment with muons giving different results? In the next decade, new measurements were made with the intention of giving an explanation for the discrepancy, but instead produced even more controversial set of results, some consistent with muonic spectroscopy and others with previously established CODATA value. This still yet unsolved problem is known as *the proton radius puzzle*.

¹Today known as the *anomalous magnetic moment* with a value of 2.79.

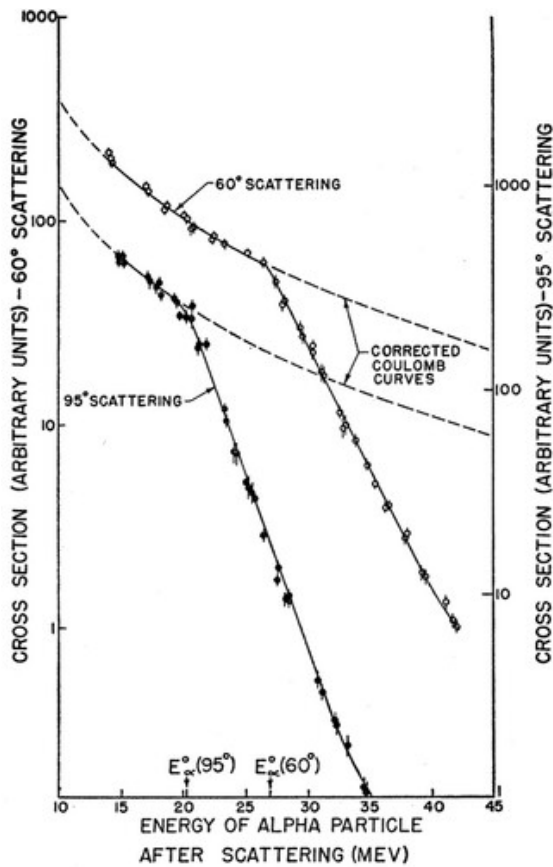


Figure 1: Cross-sections measured by Rutherford together with calculated curves, corresponding to the electric $1/r$ and magnetic $1/r^3$ potentials. The α -particles were projected on the foil with initial kinetic energies varying between 13 MeV and 42 MeV and detected at a fixed scattering angle of 60 and 95 degrees. The inverse electric potential energy coincides with the experimental points of the original figure, used as background. The Rutherford singularity appears at a kinetic energy of 28 MeV for a 60-degree angle and 20 MeV for a 95-degree angle. At higher kinetic energies, the curves deviate from the measurements due to a wrong attractive “strong force”. Figure taken from [6].

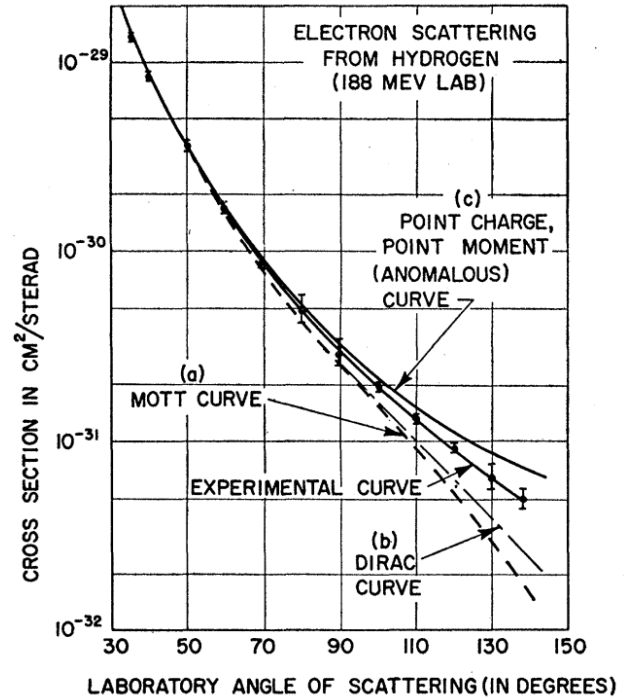


Figure 2: The deviation from the theoretical curves represent the effect of a form factor for the proton and indicates structure within the proton. Curve (a) shows the theoretical Mott curve for a spinless point proton. Curve (b) shows the theoretical curve for a point proton with the Dirac magnetic moment, and curve (c) is the theoretical curve for a point proton having the anomalous contribution in addition to the Dirac value of magnetic moment. The experimental curve falls between curves (b) and (c). Figure taken from [7].

2 Electron-proton elastic scattering

In this section, we discuss the relevance of electron-proton elastic scattering for the determination of the proton radius. We start with the first experiment that investigated electro-magnetic properties of the proton and we define the proton charge radius. Then we present some of the most important experimental results.

2.1 Differential cross-section

Rutherford formula (1) is only valid for spinless charged particles scattering off heavy nuclei, where their recoil can be neglected. Since we are dealing with light hadrons, this simplification is no longer valid because the factor of recoil becomes significant. In addition to maximize the precision of the results, alpha particle need to be replaced with a smaller and lighter probe. The obvious choice is electron, since it is the lightest charged particle without

an inner structure that predominantly interacts with electromagnetic field. With that adjustment, we are entering a relativistic regime of spin-1/2 particle that needs to be included in the theoretical description. This is done by a modified Rutherford formula, known as the *Mott cross-section*:

$$\left(\frac{d\sigma}{d\Omega}\right)_{\text{Mott}} = \left(\frac{d\sigma}{d\Omega}\right)_{\text{Ruth.}} \cdot \frac{E'}{E} \left(1 - \beta^2 \sin^2 \frac{\theta}{2}\right), \quad (2)$$

where E and E' are energies of initial and scattered electron.

But in 1956, Robert Hofstadter performed measurements of electron-proton scattering [7] and observed that experimental cross-section deviates from Mott formula, as shown in figure 2. This is due to the spatial extent of the proton's charge distribution $\rho(\mathbf{x})$. The experimental cross-section can be parametrized with additional scalar function F , such that:

$$\left(\frac{d\sigma}{d\Omega}\right)_{\text{exp.}} = \left(\frac{d\sigma}{d\Omega}\right)_{\text{Mott}} \cdot |F(\mathbf{q}^2)|^2, \quad (3)$$

where $\mathbf{q}$ is the exchanged momentum. For a particle with a charge distribution $\rho(\mathbf{x})$, function F is called the *form factor* and is defined as:

$$F(\mathbf{q}) = \int \rho(\mathbf{x}) e^{i\mathbf{q} \cdot \mathbf{x}} d\mathbf{x}. \quad (4)$$

With measurements of cross-sections at different momentum transferred we can determine charge distribution of a proton with an inverse Fourier transform of a form factor. In practice, however, the form factor can be measured only over a limited range of $|\mathbf{q}|$. The limitation is due to the finite beam energy and the rapid drop in the cross-section for large momentum transfer. Instead, we use a different approach to determine the size of a proton. By expanding the exponent (4) into Taylor series, we get:

$$\begin{aligned} F(\mathbf{q}) &= \int \left(1 + i\mathbf{q} \cdot \mathbf{x} - \frac{1}{2}(\mathbf{q} \cdot \mathbf{x})^2 - \frac{i}{3!}(\mathbf{q} \cdot \mathbf{x})^3 + \dots\right) \rho(\mathbf{x}) d\mathbf{x} \\ &= 1 - \frac{1}{6}|\mathbf{q}|^2 \langle r^2 \rangle + \dots \end{aligned} \quad (5)$$

Using the result of the expansion (5), we define the proton charge radius through the dispersion of the electric charge distribution

$$R_p^2 \equiv \langle r^2 \rangle = -6 \frac{dF(q^2)}{dq^2} \bigg|_{q^2=0}. \quad (6)$$

Since the energy loss of the electron due to the recoil is now significant, it is no longer possible to describe the interaction in terms of three-momentum transfer. Instead, the four-momentum transfer $q = (t, \mathbf{q})$, whose square q^2 is Lorentz-invariant must be used. The interpretation of the form factor as a Fourier transform of the static charge distribution is only correct for small values of $|q^2|$, since only then the three- and four-momentum transfers are approximately equal. It is also convenient to introduce $Q^2 = -q^2$ in order to work with positive quantities. The experimental precision using this approach is limited by the lowest measured value of Q^2 . In this case, parametric models are used to extrapolate values of F to $Q^2 = 0$ and then calculate derivative (6).

But taking into account that both electrons and protons are fermions, the interaction can occur not only from electric charge but also from magnetic moment. This gives rise to an additional form factor that corresponds to the distribution of magnetization inside the proton. The final form of the differential cross-section is then described by the *Rosenbluth formula*:

$$\left(\frac{d\sigma}{d\Omega}\right) = \left(\frac{d\sigma}{d\Omega}\right)_{\text{Mott}} \cdot \left[\frac{G_E^2 + \tau G_M^2}{1 + \tau} + 2\tau G_M^2 \tan^2 \frac{\theta}{2} \right], \quad (7)$$

where $\tau = Q^2/4M^2$ and M is the mass of a proton. The electric and magnetic radii can then be extracted as the slope of the proton form factors at zero momentum transfer

$$R_E^2 = -6 \frac{dG_E}{dQ^2} \bigg|_{Q^2=0}, \quad R_M^2 = -6 \frac{dG_M}{dQ^2} \bigg|_{Q^2=0}. \quad (8)$$

In the case of $Q^2 \rightarrow 0$, G_E coincides with the electric charge of the proton, normalised to the elementary charge e_0 , and G_M is equal to the magnetic moment of the proton, normalised to the nuclear magneton². To independently determine G_E and G_M , the cross-sections must be also measured at fixed values of Q^2 , for various scattering angles θ (i.e., at different beam energies E).

²Nuclear magneton is similar to Bohr magneton, corrected by a ratio of electron and nucleon mass: $\mu_N = \frac{m}{M} \mu_B$.

2.2 Measurements of electron-proton scattering cross-section

One of the most comprehensive experiments was *Bernauer et. al.* [8], performed at Johannes Gutenberg University in 2010. Their goal was to measure the cross-section of electron-proton interaction, using an electron accelerator (MAMI) with beam energies ranging from 160 MeV to 855 MeV. The design of MAMI accelerator can provide energy of the electron beam up to 1.6 GeV with intensity of $100 \mu\text{A}$.

When the electron beam reaches desired energy, it is extracted from MAMI and guided to the target and detector system, as shown in figure 3. This experiment was performed in the A1 hall, which consists of three magnetic spectrometers, that can rotate around the target to cover a wider range of the scattered angle. Design of the spectrometer with magnetic setup allows us not only to measure kinematical properties of scattered electrons with high precision but also provides information for particle identification. This is useful with higher beam energies where the inelastic scattering events are more probable. For the target, they used an aluminum cell filled with cooled hydrogen gas to reduce thermal energy of molecules and obtain lower systematical error. The final result of the experiment was $0.879(8) \text{ fm}$ for electric and $0.777(17) \text{ fm}$ for magnetic charge radius. The quality of these measurements motivated many other groups to reanalyse the data [9, 10]. Additional measurements with this spectrometer were also made in 2017 [11].

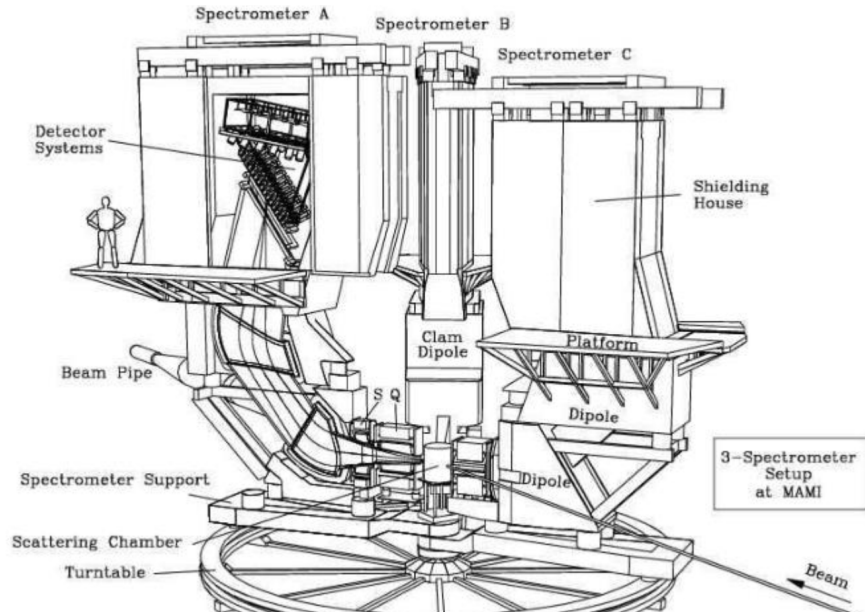


Figure 3: The three spectrometer system at Johannes Gutenberg University [12]. High-resolution magnetic spectrometers, labelled A, B, and C, can be operated in single, double, or triple coincidence mode. The electron beam comes from the bottom right and hits the target in the middle. Scattered electrons are then deflected to the vertical direction and detected in the spectrometer above.

3 Hydrogen spectroscopy

Hydrogen spectroscopy is the study of the spectral lines emitted or absorbed by hydrogen atoms. These spectral lines are the result of transitions between different energy levels of the hydrogen atom and depend on the defects of the structure. In this section, we start with a theoretical overview of hydrogen spectroscopy and then move on to the experiment that initiated the proton radius puzzle.

3.1 Theoretical background

The basic principle behind hydrogen spectroscopy is that when an electron in a hydrogen atom transitions between energy levels, it either absorbs or emits photons of specific wavelengths. These wavelengths are determined by the

energy difference between the initial and final energy levels of the electron. Hydrogen spectroscopy has been used to investigate the structure of hydrogen atoms, verify and refine theoretical models of atomic structure.

For the purposes of proton radius measurement, it is important to consider that the energy levels of a hydrogen atom include corrections coming from QED, hyper-fine splitting and additionally the finite size of a proton. The goal is to measure energy shifts caused by these effects, compare them to theoretical calculations, and extract the proton charge radius. All the corrections also include the Rydberg constant, which is one of the most precisely measured constants in physics.

The first approximation of bound state energies can be obtained from the Schroedinger equation using Coluomb potential, which is known as the *Schroedinger model*:

$$E_n = -\frac{mc^2}{2n^2}\alpha^2 = -hc\frac{R_\infty}{n^2}, \quad (9)$$

where n is the principal quantum number, m is the mass of the electron, and R_∞ is the Rydberg constant. Each state is additionally labelled with orbital angular momentum l and its projection m_l . Therefore, n -th energy level is n times degenerate with respect to l and each l is $2l + 1$ times degenerate with respect to m_l . If we take into account finite mass of the centre nuclei, the reduced mass m_r has to be introduced as

$$m_r = \frac{mM}{m + M} \quad (10)$$

and the corrected energy levels are then written as m_r/mE_n .

However, better results are derived from the *Dirac model*, which manifestly eliminates degeneracy of the energy states with $n \geq 2$. Description with relativistic correction of a spin-1/2 particle is given by Dirac equation with a Coluomb source with finite mass

$$E_{nj} = Mc^2 + m_rc^2(f(n, j) - 1), \quad (11)$$

where

$$f(n, j) = \left(1 + \frac{\alpha^2}{(n - j - \frac{1}{2} + ((j + 1/2)^2 - \alpha^2)^2)}\right)^{-\frac{1}{2}} \approx 1 - \frac{\alpha^2}{2n^2} - \frac{\alpha^4}{2n^3} \left(\frac{1}{j + 1/2} - \frac{3}{4n}\right) + \dots \quad (12)$$

But the experimental accuracy of spectroscopy demands other corrections, like hyperfine splitting (HFS) due to spin-spin interaction, Lamb shift due to vacuum polarization and finally correction due to the finite volume of the nucleus. The Lamb shift can be expressed as:

$$\Delta E_{nlj}^{Lamb} = \frac{4mc^2}{3\pi n^3} \alpha^5 \left[L_{nl} + [19/30 - 2 \ln \alpha] \delta_{l0} \pm \frac{3}{4} \frac{1}{(2j + 1)(2l + 1)} (1 - \delta_{l0}) \right], \quad (13)$$

where L_{nl} is a numerically computed term, called *Bethe logarithm*. Hyperfine splitting, on the other hand, depends on the angular momentum of nucleus $\mathbf{I}$ and electron $\mathbf{J}$, which can be combined to give $\mathbf{F} = \mathbf{I} + \mathbf{J}$. Energy contribution is then proportional to $\Delta E_{nljIF} \propto [F(F + 1) - I(I + 1) - j(j + 1)]$. The finite size contribution is significant only for S states. A physical interpretation of this effect can also be obtained from radial wave functions, where only S states overlap with proton in a way that a difference can be measured. Since the proton charge is distributed over a finite volume, the electron experiences smaller attraction, leading to energy correction

$$\Delta E_V = -\frac{2}{3} \frac{\alpha^4}{n^3} m_r^3 c^6 \left(\frac{R_p}{\hbar c}\right)^2 \delta_{l0} + \mathcal{O}(R_p^3), \quad (14)$$

from which the proton charge radius can be extracted. All detailed derivations on corrections, including proof that R_p corresponds to the definition (8), can be found in [13, 14].

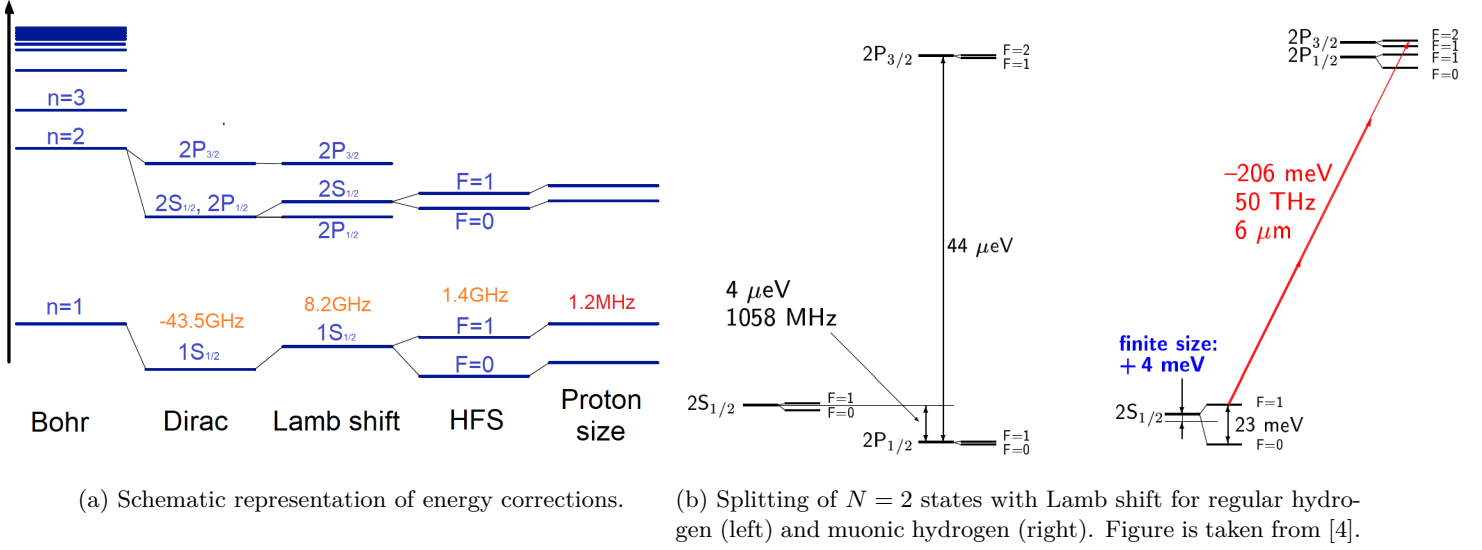


Figure 4: Schematic representation of energy levels in hydrogen spectroscopy.

3.2 Measurements of Lamb shift

Since the first measurement of the proton charge radius with hydrogen spectroscopy in 1979, over 15 experiments were made [2], including $2S$ - $2P$, $2S$ - $2P$, $2P$ - $4D$ and other transitions. Until 2010, the average value of the proton radius, obtained from hydrogen spectroscopy was $0.8779(94)\text{ fm}$ [15].

Muonic hydrogen spectroscopy is similar to hydrogen spectroscopy, with the exception that an electron is replaced with a muon. As shown in equations (13) and (14), energy corrections depend on the electron's mass. Therefore, if an electron is replaced by a 205-times heavier muon, the difference between energy states changes significantly (figure 4b) and the contribution of finite size (14) is increased by a factor of $(m_\mu/m_e)^3 \approx 6 \cdot 10^6$. As a result, the extraction of the proton charge radius can be done much more precisely.

The new experiment [4] was performed at the Paul Scherrer Institute (PSI), Switzerland. It was based on the measurement of the energy difference between the $2S_{1/2}^{F=1}$ and $2P_{1/2}^{F=2}$ muonic energy levels. This transition was chosen because it gives the largest signal of all six allowed optical $2S - 2P$ muonic transitions. The first step was to produce pions by inelastic scattering with high-energy protons. Pions then decay into muons ($\pi \rightarrow \mu + \bar{\nu}_\mu$) by the weak interaction. The muon beam is then slowed down and guided through a 5T solenoid, where individual muons are detected to provide a laser trigger. At the end of the solenoid is placed a cell with ordinary hydrogen, where muons are “captured”, usually in one of the outer energy levels $n \approx 14$. By a cascade of radiative transitions, about 99% of the muons proceed to the $1S$ ground state, emitting X-rays, while the remaining 1% populate the metastable $2S$ state. But the presence of atom-to-atom collisions causes $2S$ state to almost immediately decay into the ground state, so the pressure of the hydrogen cell has to be reduced to 10^{-3} bar . Triggered by the detection of an incoming muon, a laser pulse illuminates the muon in order to make the transition. The transition signal (figure 5) is obtained by detecting the photons emitted after a successful laser-induced $2S$ to $2P$ transition. A signal rate at the peak of the resonance of only about six detected photons per hour was achieved after offline cuts. The measurement was possible due to the small background rate of one event per hour. Several hundred laser-induced events were detected per resonance (figure 6), leading to a statistical uncertainty of a few percent of the width. In this experiment, it was obtained that the frequency of the laser to induce the $S_{1/2}^{F=1} - 2P_{3/2}^{F=2}$ transition corresponds to an energy of $\Delta E = 206.2949(32)\text{ meV}$. Using detailed calculations of QED, HFS, and size corrections, the total energy difference of this transition was obtained as

$$\Delta E [\text{meV}] = 209.9779(49) - 5.2262R_p^2 + 0.0347R_p^3, \quad (15)$$

from which the proton charge radius was extracted with a value of $0.8418(4)\text{ fm}$. This new value of the proton radius was nearly ten times more precise than the previous world average from hydrogen spectroscopy and electron-proton scattering, but seven standard deviations smaller.

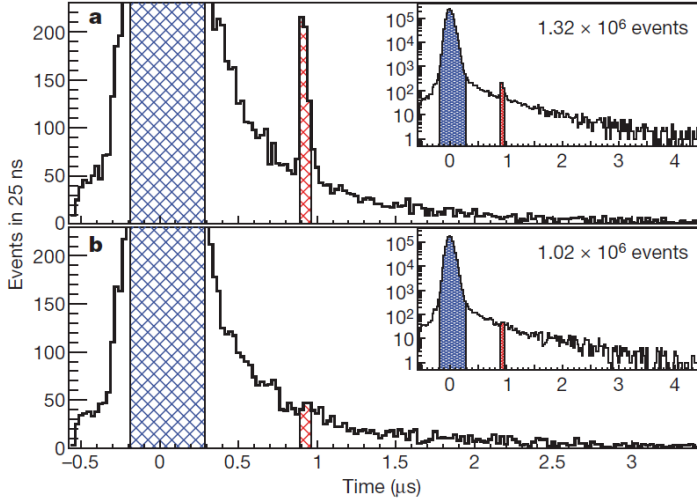


Figure 5: X-ray spectra recorded on resonance (upper fig.) and off resonance (lower fig.). Blue-colored peak represents a cascade of transitions (“prompt” X-rays) to the ground state right after muon capture. Red reign are photons emitted with deexcitation to the ground state after a successful $2S - 2P$ transition. Figure taken from [4].

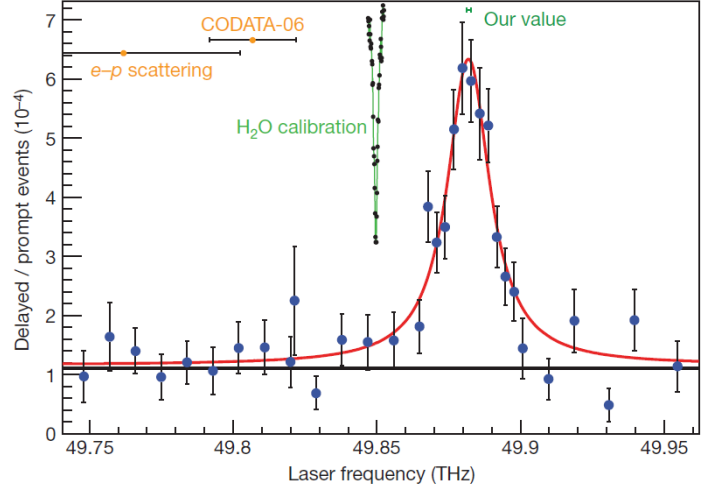


Figure 6: Resonance curve matching number of events in the laser time window normalized to the number of “prompt” events as a function of the laser frequency. For comparison, previous results of electron-proton scattering and CODATA 2006 are converted into units of photon frequency and marked yellow. One of the calibration measurements using water absorption is also shown (green line). Figure is taken from [4].

To escalate the proton radius puzzle, the second experiment [5] was performed in 2013. In order to rule out some possible errors arising from theoretical predictions, two energy transitions were measured. In the new experiment, additional $2S_{1/2}^{F=0} \rightarrow 2P_{3/2}^{F=1}$ transition was measured as well as the previously studied one, using the same technique. The measurement of the two transitions allowed for an extraction of the proton radius without relying on the theoretical predictions of the hyperfine splitting of the $2S$ muonic energy level, and only theoretical predictions of the HFS for the $2P$ energy levels were still needed. The new value of $0.84087(39)$ fm obtained is 1.7 times more precise than the previous measurement of 2010 with muonic hydrogen and in agreement with it, but still in strong disagreement with the CODATA value of $0.8770(45)$ fm.

To escalate things even further, in 2017 and 2019, two new experiments [15, 16] took place using regular hydrogen spectroscopy. With improved experimental setup and reduced systematics, they measured a proton radius of $0.8335(95)$ fm and $0.833(10)$ fm. The results of newer experiments happened to be in favour of a smaller proton radius, and therefore the CODATA published a recommended value of 0.8414 fm in 2018.

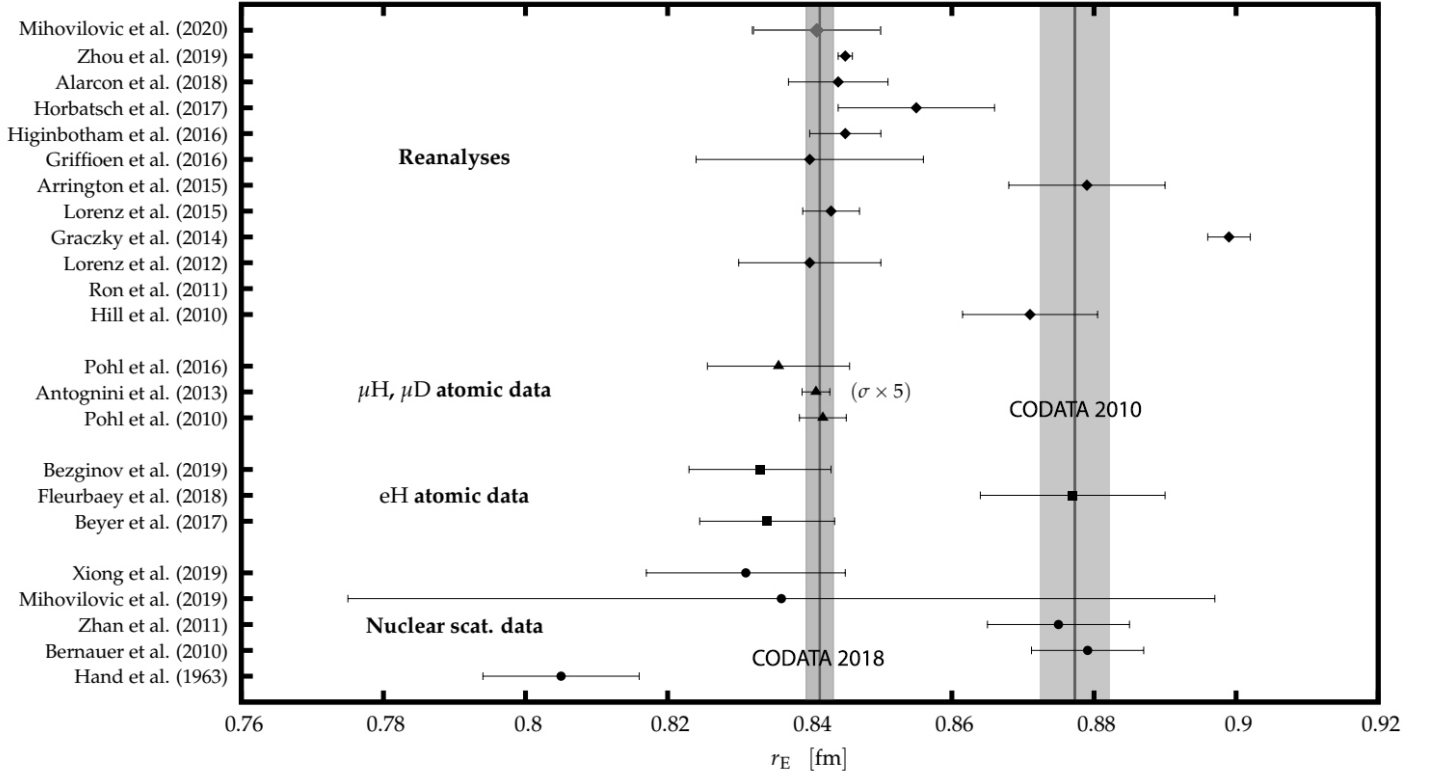


Figure 7: Comparison of the most relevant contributions to the proton radius, divided into four groups: rhombuses at the top represent reanalyses of [8], triangles represent measurements of muonic hydrogen and deuterium measurements, squares represent regular hydrogen spectroscopy, and circles represent elastic electron-proton scattering. The two gray bands represent the values of CODATA 2010 and 2018. The picture is taken from [9].

4 Possible solutions

Since 2010, a lot of effort has been made trying to explain the discrepancy of the measurements, and many explanations have been proposed, some claiming that the calculations that are used in the spectroscopy are incomplete - there are additional terms that need to be taken into account because of the high precision of the measurements [17] such as proton polarization of higher order or that higher moments of charge distribution need to be taken into account for the extraction of R_p . But many additional terms are calculated in [18, 19] and found that this could not be the reason.

Other explanations rely on problems with extrapolation of scattering data at low Q^2 . Since the data provided by the experiment does not cover a close range of the point where R_p is defined, extrapolation is necessary, and by choosing different models, the final results can vary. This model dependency can also be seen from the results of reanalyses of Bernauer measurements in figure 7 and is also described in [12].

Additional arguments were proposed regarding incompleteness of the Standard model. Beyond standard model (BSM) theories imply that the lepton universality is violated and new force carriers exist [20]. The theory of new gauge particles could simultaneously explain the muonic $g - 2$ puzzle as well. But none of the proposed explanations could resolve the proton radius puzzle without creating additional issues with consistency with other established theories. Despite numerous new attempts and reanalyses, no significant breakthrough has been made. Progress is crucially dependent on conducting new, more precise experiments. One of them is to measure the scattering cross-section in muon-proton scattering, which is being done by the Muonic Scattering Experiment (MUSE). Its goal is to simultaneously measure scattering cross-sections of electrons and muons, providing insights into potential differences when scattering different lepton flavours off hydrogen.

5 Conclusion

After more than a decade of effort, the proton puzzle, sparked unexpectedly in 2010 by Bernauer [8] and Pohl [4], still remains unsolved. New results from the measurement of the Lamb shift in electronic hydrogen [16] and an upgraded elastic scattering experiment [21] have dispelled some doubts and established a new, smaller value for the proton charge radius. However, despite this, some details remain unresolved, such as why certain results from the data analysis of the 2010 scattering experiment, to which the obtained result at 0.876(6) fm and 0.868(48) fm are attributed, and even one of the recent attempts to measure the Lamb shift, still favour a larger value of the proton radius. This issue has received considerable attention, and there appears to be no clear answer. Therefore, for the final explanation of the puzzle of the proton radius, it will be necessary to wait for further results.

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